Lesson 11: Variance

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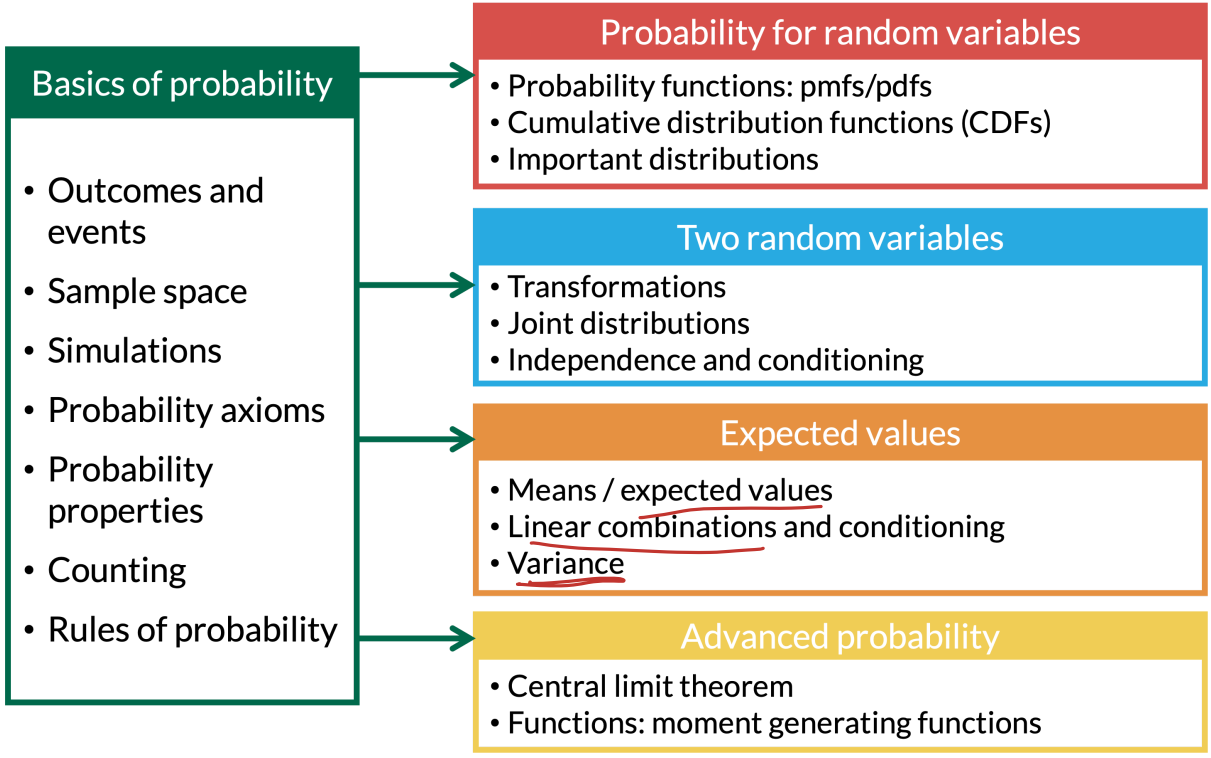
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Learning Objectives

1. Define and calculate the expected value for a function of discrete and continuous RVs
2. Define and calculate variance for a single random variable
3. Define and calculate variance for multiple random variables

Where are we?


Basics of probability
• Outcomes and events
• Sample space
• Simulations
• Probability axioms
• Probability properties
• Counting
• Rules of probability
Probability for random variables
• Probability functions: pmfs/pdfs
• Cumulative distribution functions (CDFs)
• Important distributions
Two random variables
• Transformations
• Joint distributions
• Independence and conditioning
Expected values
• Means / expected values
• Linear combinations and conditioning
• Variance
Advanced probability
• Central limit theorem
• Functions: moment generating functions

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Let's start building the variance through expected values of functions

Example 1

Let g be a function and let $g(x) = ax + b$, for real-valued constants a and b. What is $\mathbb{E}[g(X)]$?

$$\begin{aligned} E[g(X)] &= E[aX + b] \\ &= E(aX) + E(b) \\ &= a\,E(X) + b \end{aligned}$$

$g(X)$ is linear fn w/ constants

What if $g(X) = X^2 = X \cdot X$

$$E(g(X)) = E(X^2) \neq E(X) \cdot E(X)$$

$$\neq [E(X)]^2$$

need a def'n of exp val for non-lin fn $g(X)$

What is the expected value of a function?

Expected value of function of discrete RV

For any function g and discrete RV X, the expected value of $g(X)$ is

$$\mathbb{E}[g(X)] = \sum_{\{all\ x\}} g(x) p_X(x)$$

Expected value of function of continuous RV

For any function g and continuous RV X, the expected value of $g(X)$ is

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

- For example, if we have $g(x) = x^2$, then

$$\mathbb{E}[X^2] = \sum_{\{all\ x\}} x^2 p_X(x) \neq \left(\sum_{\{all\ x\}} x p_X(x) \right)^2 = (\mathbb{E}[X])^2$$

value, weight, weighted avg

binom (n = 2, p)

$E(X^2) = 0^2 p_X(x=0) + 1^2 p_X(x=1) + 2^2 p(x=2)$

$(E(X))^2 = (0 \cdot p(X=0) + 1\, p(X=1) + 2\, p(X=2))^2$

Let's revisit the card example (1/2)

Example 2

Suppose you draw 1 card from a standard deck of cards. Let $X = 1$ if the card is a heart and $X = 0$ if it is not.

1. Find $\mathbb{E}[X^2]$.

Recall Bernoulli RV (Lesson 10, Example 3):

$$X = \begin{cases} 1 & \text{with probability } p \\ 0 & \text{with probability } 1-p \end{cases}$$

$\mathbb{E}[X] = p$. Here, $p = \mathbb{P}(\text{heart}) = \frac{1}{4}$

$$E(g(X)) = E(X^2)$$

$$= \sum_{\{all\ x\}} x^2 \, p(X = x)$$

$$= 0^2 \, P(X=0) + 1^2 \, P(X=1)$$

$$= 1 \cdot p = p$$

$P(X=1) = p$

$$E(X^2) \neq [E(X)]^2$$

$$= p \qquad p^2$$

Let's revisit the card example (2/2)

Example 2

Suppose you draw 1 card from a standard deck of cards. Let $X = 1$ if the card is a heart and $X = 0$ if it is not.

2. Find $\mathbb{E}\left[\left(X - \frac{1}{4}\right)^2\right]$.

Recall Bernoulli RV (Lesson 10, Example 3):

$$X = \begin{cases} 1 & \text{with probability } p \\ 0 & \text{with probability } 1-p \end{cases}$$

$\mathbb{E}[X] = p$. Here, $p = \mathbb{P}(\text{heart}) = \frac{1}{4}$

With $g(x) = \left(x - \frac{1}{4}\right)^2$ and $p = \frac{1}{4}$:

$$\mathbb{E}\left[\left(X - \frac{1}{4}\right)^2\right] = \sum_{\{all\ x\}} \left(x - \frac{1}{4}\right)^2 p_X(x)$$

$$= \left(0 - \frac{1}{4}\right)^2 (1-p) + \left(1 - \frac{1}{4}\right)^2 p$$

$$= \frac{1}{16} \cdot \frac{3}{4} + \frac{9}{16} \cdot \frac{1}{4} = \frac{3}{64} + \frac{9}{64} = \frac{12}{64} = \frac{3}{16}$$

Another way:

$$E\left[\left(X - \tfrac{1}{4}\right)^2\right] = E\left[\left(X - \tfrac{1}{4}\right)\left(X - \tfrac{1}{4}\right)\right]$$

$$= E\left[X^2 - \tfrac{1}{2}X + \tfrac{1}{16}\right]$$

$$= E(X^2) - E\left(\tfrac{1}{2}X\right) + E\left(\tfrac{1}{16}\right)$$

prev slide: $E(X^2) = \frac{1}{4}$; prev class: $-\frac{1}{2}E(X) = -\frac{1}{2}\left(\frac{1}{4}\right)$; $E\left(\frac{1}{16}\right) = \frac{1}{16}$

$$= \frac{1}{4} - \frac{1}{2}\left(\frac{1}{4}\right) + \frac{1}{16} = \frac{4 - 2 + 1}{16} = \frac{3}{16}$$

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Variance of a RV

Definition: Variance of RV

(units of X^2)

The **variance** of a RV X, with (finite) expected value $\mu_X = \mathbb{E}[X]$ is

for variance: $g(X) = (X - \mu_X)^2$

$$\sigma_X^2 = Var(X) = \mathbb{E}[(X - \mu_X)^2] = \mathbb{E}[(X - \mathbb{E}[X])^2]$$

Definition: Standard deviation of RV

The **standard deviation** of a RV X is

$$\sigma_X = SD(X) = \sqrt{\sigma_X^2} = \sqrt{Var(X)}.$$

Variance of discrete and continuous RVs

How do we calculate the variance of a discrete RV?

For discrete RVs:

$$
\begin{aligned}
Var(X) &= \sum_{\{all\ x\}} (x-\mu_X)^2 p_X(x) \\
&= \mathbb{E}[(X-\mu_X)^2] \\
&= \mathbb{E}[(X-\mathbb{E}[X])^2] \\
&= \mathbb{E}[X^2] - (\mathbb{E}[X])^2
\end{aligned}
$$

How do we calculate the variance of a continuous RV?

For continuous RVs:

$$
\begin{aligned}
Var(X) &= \int_{-\infty}^{\infty} (x-\mu_X)^2 f_X(x)dx \\
&= \mathbb{E}[(X-\mu_X)^2] \\
&= \mathbb{E}[(X-\mathbb{E}[X])^2] \\
&= \mathbb{E}[X^2] - (\mathbb{E}[X])^2
\end{aligned}
$$

Let's calculate the variance and prove it!

Lemma 6: "Computation formula" for Variance

The variance of a RV X, can be computed as

$$\begin{aligned}\sigma_X^2 &= Var(X) \\ &= \mathbb{E}[X^2] - \mu_X^2 \\ &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2\end{aligned}$$

Two facts from Lesson 10:

- $\mu_X = \mathbb{E}[X]$ is a constant
- $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$

Plan: start from the definition, expand the square, and use what we know about expected values

$$\begin{aligned}Var(X) &= \mathbb{E}[(X - \mu_X)^2] && \text{definition of variance} \\ &= \mathbb{E}[X^2 - 2\mu_X X + \mu_X^2] && \text{expand the square} \\ &= \mathbb{E}[X^2] - 2\mu_X\mathbb{E}[X] + \mu_X^2 && \text{split the sum; } \mu_X \text{ is constant} \\ &= \mathbb{E}[X^2] - 2\mu_X^2 + \mu_X^2 && \mathbb{E}[X] = \mu_X \\ &= \mathbb{E}[X^2] - \mu_X^2 && \text{combine the } \mu_X^2 \text{ terms}\end{aligned}$$

linear comb

$E(X^2) + E(-2\mu_X X) + E(\mu_X^2)$

$E(X^2) - [E(X)]^2$ → exp val of X^2 − sq of exp val

- This works for both discrete and continuous RVs!
- Want to see it written out with sums for a discrete RV? See the muddy points for this lesson.

Let's simulate the card example (1/2)

- Same steps as Lesson 2's dice, now with `n_cards <- 1`
- Each suit is a quarter of the deck, so each suit is equally likely

```
1  set.seed(1028)                                                          ①
2  reps <- 10000                                                           ②
3  n_cards <- 1
4  suits <- c("hearts", "diamonds", "clubs", "spades")                     ③
5
6  draws <- expand_grid(sim = 1:reps, card = 1:n_cards) |>                 ④
7    mutate(suit = sample(suits, n(), replace = TRUE))
8
9  X_sims <- draws |>                                                      ⑤
10   group_by(sim) |>
11   summarise(X = sum(suit == "hearts"))
```

- The first 20 simulated values of X:

```
1  head(X_sims$X, 20)                                                      ①
```

```
 [1] 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 0
```

- X can only be 0 or 1, just like a Bernoulli RV

Let's simulate the card example (2/2)

- Let's calculate some of the values we've shown by hand

```
X_sims |>
  summarise(
    mean_X        = mean(X),
    mean_X_sq     = mean(X^2),
    variance      = mean((X - mean(X))^2),
    var_shortcut  = mean(X^2) - mean(X)^2,
    std_dev       = sqrt(variance)
  )
```

```
# A tibble: 1 × 5
  mean_X mean_X_sq variance var_shortcut std_dev
   <dbl>     <dbl>    <dbl>        <dbl>   <dbl>
1  0.249     0.249    0.187        0.187   0.432
```

- Both formulas for the variance give the same number!

var() > sd() preset fns in R

Exact answers for the card example:

- $\mathbb{E}[X] = \frac{1}{4}$
- $\mathbb{E}[X^2] = \frac{1}{4} = 0.25$
- $Var(X) = \frac{1}{4} - \left(\frac{1}{4}\right)^2 = \frac{3}{16} = 0.1875$
- $SD(X) = \sqrt{3/16} \approx 0.433$

Variance of a continuous Uniform distribution

Example 3

Let $f_X(x) = \frac{1}{b-a}$, for $a \le x \le b$. Find $Var[X]$.

last class: $E(X) = \frac{a+b}{2}$

$$E(X^2) = \int_a^b x^2 \frac{1}{b-a}\,dx \qquad g(X) = x^2$$

$$= \frac{1}{3(b-a)} x^3 \Big|_{x=a}^{x=b}$$

$$= \frac{1}{3(b-a)}\left(b^3 - a^3\right)$$

$$= \frac{b^2 + ab + a^2}{3}$$

do alg @ Home if you want to see

$$Var(X) = E(X^2) - [E(X)]^2$$

solve on side

$$= \frac{b^2+ab+a^2}{3} - \left(\frac{a+b}{2}\right)^2$$

$$= \frac{4(b^2+ab+a^2) - 3(a+b)^2}{12}$$

$$= \frac{4b^2 + 4ab + 4a^2 - 3a^2 - 6ab - 3b^2}{12}$$

$$= \frac{b^2 - 2ab + a^2}{12} = \frac{(b-a)^2}{12}$$

$$\boxed{Var(X) = \frac{(b-a)^2}{12}}$$

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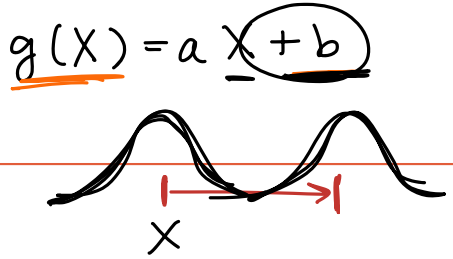
Variance of a function with a single RV

Lemma 7

For a RV X and constants a and b,

$$Var(aX + b) = a^2 Var(X).$$

Proof will be exercise in homework. It's fun! In a mathy kinda way.



spread of data stay the same w/ shifts like b (variance is a meas of spread)

Important results for *independent* RVs

$\mathbb{E}(X^2) \neq [\mathbb{E}(X)]^2$

Theorem 8

For independent RV's X and Y, and functions g and h,

$$\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\mathbb{E}[h(Y)].$$

Corollary 1

For independent RV's X and Y,

$\mathbb{E}(X \cdot X)$

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y].$$

Why Theorem 8 works (discrete example). Independence means $p_{X,Y}(x,y) = p_X(x)p_Y(y)$, just like $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$:

$$\begin{aligned}
\mathbb{E}[g(X)h(Y)] &= \sum_{\{all\ x\}} \sum_{\{all\ y\}} g(x)h(y)\, p_{X,Y}(x,y) && \text{expected value of a function} \\
&= \sum_{\{all\ x\}} \sum_{\{all\ y\}} g(x)h(y)\, p_X(x)\, p_Y(y) && \text{use independence} \\
&= \left(\sum_{\{all\ x\}} g(x)p_X(x) \right) \left(\sum_{\{all\ y\}} h(y)p_Y(y) \right) && \text{the } x \text{ and } y \text{ parts separate} \\
&= \mathbb{E}[g(X)]\,\mathbb{E}[h(Y)]
\end{aligned}$$

Variance of sum of independent RVs

exp val: linearity (linear combos of RVs), regardless of independence

Theorem 9: Variance of sum of independent RVs

For independent RVs (discrete or continuous) X_i and constants $a_i, i = 1, 2, \dots, n$,

$$Var\left(\sum_{i=1}^{n} a_i X_i\right) = \sum_{i=1}^{n} a_i^2 Var(X_i).$$

Simpler version: X & Y are independent

$$Var(a_1X + a_2Y) = Var(a_1X) + Var(a_2Y) = a_1^2 Var(X) + a_2^2 Var(Y)$$

linearity of variance for independent RVs

Corollaries

Corollary 2

For independent RVs $X_i, i = 1, 2, ..., n,$

$$Var\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} Var(X_i).$$

Corollary 3

For independent identically distributed (i.i.d.) RVs $X_i, i = 1, 2, ..., n,$

$$Var\left(\sum_{i=1}^{n} X_i\right) = nVar(X_1).$$

Ghost problem: *with replacement*

X = # chocolates (w/ rep)

~~Var(X) E(X²) [E(X)]²~~

$X = \sum_{i=1}^{5} Y_i$ $\quad Y_i \overset{iid}{\sim} \text{Bernoulli}\left(p = \frac{1}{6}\right)$

Example 5

The ghost is trick-or-treating at a ~~different~~ house ~~again~~. In this case it is known that the bag of candy has 10 chocolates, 20 lollipops, and 30 laffy taffies. The ghost grabs a handful of five pieces of candy. What is the variance for the number of chocolates the ghost takes? Let's solve this for the case ***with*** replacement.

$p = \frac{1}{6}$ for choco

FOR 1 CANDY:

Recall probability for Bernoulli:

$$p_Y(y) = p^y(1-p)^{1-y} \text{ for } y = 0, 1$$

in L10

From Example 2: $\mathbb{E}[Y] = \mathbb{E}[Y^2] = p$

$$\text{Var}(X) = \text{Var}\left(\sum_{i=1}^{5} Y_i\right) = \sum_{i=1}^{5} \text{Var}(Y_i)$$

b/c ind

$$= 5\,\text{Var}(Y_1)$$

b/c iid

$$\text{Var}(Y_1) = E(Y_1^2) - [E(Y_1)]^2$$

$E(Y_1^2) = p$, $E(Y_1) = p$

$$= p - (p)^2 = p(1-p)$$

$$= 5\left(p(1-p)\right) = 5\left(\frac{1}{6}\right)\left(1 - \frac{1}{6}\right)$$

$$= 0.6944$$

Var of binom: n · var of bern

Back to our hotel example from Lesson 10

Example 6

A tour group is planning a visit to the city of Minneapolis and needs to book 30 hotel rooms. The average price of a room is $200 **with standard deviation $10**. In addition, there is a 10% tourism tax for each room. What is the **standard deviation** of the cost for the 30 hotel rooms? Assume rooms are independent.

Problem to do at home if we don't have enough time.

Set up: Let T = total cost of 30 rooms
C_i = cost of room i (pre tax)

$$T = \sum_{i=1}^{30} 1.1\, C_i$$

Want to find SD of T (start w/ Var(T))

$$\text{Var}(T) = \text{Var}\left[\sum_{i=1}^{30} 1.1\, C_i\right] = \text{Var}\left[1.1 \sum_{i=1}^{30} C_i\right]$$

$$= 1.1^2\, \text{Var}\left[\sum_{i=1}^{30} C_i\right] = 1.1^2 \sum_{i=1}^{30} \text{Var}(C_i)$$

b/c ind rooms

b/c all Var(C_i) same

$$= 1.1^2 \cdot 30\, \text{var}(C_1) = 1.1^2 \cdot 30 \left[\text{SD}(C_i)\right]^2$$

$$= 1.1^2 \cdot 30 \cdot (10^2) = \$^2\, 3{,}600$$

$$\text{SD}(T) = \sqrt{\text{var}(T)} = \sqrt{\$^2 3{,}600} = \$60$$

The standard deviation of the cost of rooms is $60.

WRONG TO DO: $T = 30 \cdot 1.1 \cdot C_i$

$\text{Var}(T) = \text{var}(30 \cdot 1.1 \cdot C_i)$

$= (30 \cdot 1.1)^2\, \text{Var}(C_i)$

Find the mean and sd from word problem (1/3)

Example 7

A machine manufactures cubes with a side length that varies uniformly from 1 to 2 inches. Assume the sides of the base and height are equal. The cost to make a cube is 10 ¢ per cubic inch and 5 ¢ for the general cost per cube. Find the mean and standard deviation of the cost to make 10 cubes.

unif dist

Side work: $f_X(x) = 1$ for $1 \le x \le 2$

$$\mathbb{E}[X_i^3] = \int_1^2 x^3 \cdot 1 \, dx = \frac{x^4}{4}\Big|_1^2 = \frac{16-1}{4} = \frac{15}{4}$$

$$\mathbb{E}[X_i^6] = \int_1^2 x^6 \cdot 1 \, dx = \frac{x^7}{7}\Big|_1^2 = \frac{128-1}{7} = \frac{127}{7}$$

Set up:

let C = cost of 10 cubes (in cents)

C_i = cost of cube i $(i = 1, 2, \ldots, 10)$

X_i = length of each side of cube i

$$C = \sum_{i=1}^{10} C_i \qquad C_i = 5 + 10(X_i^3)$$ → def of volume of cube

MEAN

$$E(C) = E\left[\sum_{i=1}^{10} C_i\right] = E\left[\sum_{i=1}^{10} (5 + 10(X_i^3))\right]$$

$$= \sum_{i=1}^{10} E(5 + 10X_i^3) = \sum_{i=1}^{10} [E(5) + 10E(X_i^3)]$$

linearity; 5 & 10 constants

$$= \sum_{i=1}^{10} \left(5 + 10\left(\frac{15}{4}\right)\right) = \sum_{i=1}^{10} 42.5 = 425\text{¢}$$

$4.25

mean cost of 10 cubes is $4.25

Find the mean and sd from word problem (2/3)

$g(X) = b + aX^3$

$\text{Var}(b + aX^3) = a^2 \text{Var}(X^3)$

$$\text{Var}(C) = \text{Var}\left[\sum_{i=1}^{10}(5 + 10X_i^3)\right] = \sum_{i=1}^{10} \text{Var}(5 + 10X_i^3)$$

each cube (C_i) is independent

$$= \sum_{i=1}^{10} 10^2 \, \text{Var}(X_i^3)$$

$$= 10^2 \sum_{i=1}^{10} \text{Var}(X_i^3)$$

$$= 100 \sum_{i=1}^{10} 4.0803$$

$$= 100 \cdot 10 \cdot 4.0803$$

$$= 4080.36 \text{¢}^2$$

$$\text{Var}(X_i^3) = E[(X_i^3)^2] - [E(X_i^3)]^2$$

$(X_i^3)^2 = X_i^6$

$$= \frac{127}{7} - \left(\frac{15}{4}\right)^2$$

$$= 4.0803$$

$$sd(C) = \sqrt{4080.36} = 63.8776\text{¢} = \boxed{64\text{¢}}$$

Find the mean and sd from word problem (3/3)

Let's simulate the cost of 10 cubes: simulate

- Same steps as the card simulation, with `runif()` from Lesson 8 for the Uniform(1, 2) side lengths

```
1  reps <- 10000                                          ①
2  n_cubes <- 10
3
4  cubes <- expand_grid(                                  ②
5    sim = 1:reps,
6    cube = 1:n_cubes
7    ) |>
8    mutate(
9      side = runif(n(), min = 1, max = 2),               ③
10     cost = 5 + 10 * side^3                             ④
11     )
12
13 cost_sims <- cubes |>                                  ⑤
14   group_by(sim) |>
15   summarise(total_cost = sum(cost))
```

```
1  cubes
```

```
# A tibble: 100,000 × 4
     sim  cube  side  cost
   <int> <int> <dbl> <dbl>
 1     1     1  1.47  36.5
 2     1     2  1.45  35.5
 3     1     3  1.10  18.4
 4     1     4  1.99  84.3
 5     1     5  1.88  71.4
 6     1     6  1.78  61.3
 7     1     7  1.09  17.9
 8     1     8  1.10  18.4
 9     1     9  1.41  33.2
10     1    10  1.67  52.0
# ℹ 99,990 more rows
```

Let's simulate the cost of 10 cubes: summarize

- Mean and SD of the total cost:

```
cost_sims |>
  summarise(
    average = mean(total_cost),
    std_dev = sd(total_cost)
  )
```

```
# A tibble: 1 × 2
  average std_dev
    <dbl>   <dbl>
1    425.    63.7
```

From Example 7: $\mathbb{E}[C] = 425¢$ and $SD(C) = \sqrt{4080.36} \approx 63.9¢$